

Daksh Agrawal

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Summary

AI Engineer specializing in **Generative AI** systems, autonomous **Agentic AI workflows**, and intelligent automation. Proven expertise in **LLM orchestration**, custom **tool-calling mechanisms**, and production-grade **RAG architectures** integrated with high-performance data engines. Thrives in highly dynamic, fast-paced environments — with a track record spanning elite venture capital (**Square Peg**, \$3.8B AUM), critical utilities (**TasWater**), and high-growth start-ups (**G7 Smart Logistics**). A systems-first builder adept at translating complex historical artifacts and massive unstructured datasets into production-ready AI solutions with robust **evaluation guardrails**, focused on reducing operational latency and shipping fluid end-to-end applications.

Analytical & Professional Skills

- **Languages:** Python, SQL, JavaScript/TypeScript, Java, Kotlin
- **AI-Assisted Development:** Context Engineering, Cursor, Claude Code, AI-Generated Code Review & Validation
- **Frameworks & Tools:** Next.js, FastAPI, n8n, LangChain, Git, PostgreSQL
- **Data & AI Ecosystem:** Pandas, Polars, RAG, NumPy, OpenAI API, LLM Agents, Power Query, PyTorch, scikit-learn
- **DevOps & Cloud:** AWS, Cloudflare, Docker, GitHub Actions, CI/CD, MLOps, Observability
- **Machine Learning & Statistical Modelling:** A/B testing, Time-series forecasting, Financial Modelling

Work Experience

Data Scientist (Graduate Program)

Feb 2026 — Present

TasWater

Hobart, TAS

- **Dashboard Development:** Built and deployed end-to-end Power BI solutions for KPI tracking, leveraging SQL and Power Query for efficient data warehousing and modeling.
- **Process Automation:** Spearheaded automation initiatives that reduced reporting turnaround time by 4 days and saved 200+ man-hours annually.
- **Operational Analysis:** Enhanced execution and accountability within large-scale public-sector operations by providing structured analysis for the Supply Chain and Project Management teams.
- **Cross-Functional Collaboration:** Acted as a bridge between technical data teams and business units to ensure reporting aligned with strategic organizational goals.

Data & AI Intern

Mar 2025 — Jan 2026

Square Peg -Global Tech Fund

Melbourne, VIC

- **Engineered Portfolio Intelligence Pipeline:** Built a scalable LinkedIn scraper to ingest **500k+ professional profiles** per day to Snowflake Warehouse; integrated LLM-driven enrichment pipelines to generate automated, real-time insights on hiring velocities and departure trends across portfolio companies.
- **Developed RAG-Powered Podcast Search:** Processed and Transcribed **8,000+ hours of audio content** (podcasts) into a vector database to power an internal **Retrieval-Augmented Generation (RAG) search system**, enabling investment analysts to instantly retrieve historical context paired with accurate, verifiable citations.
- **Automated Investment Analytics Pack Generator:** Designed and built a Claude-powered generation engine that ingested historical analyst artifacts to replicate multi-company deep dives, **automatically rendering quantitative metric packages and interactive data visualizations**; accelerated high-fidelity presentation pack generation from **20+ hours down to 5–6 minutes**.
- **Created “Dakshboards” Analytics Framework:** Architected a predictive ML valuation model and a fund performance reporting framework handling highly complex financial data; built minimalist, high-adoption internal dashboards (“Dakshboards”) that became the organization’s primary interface for portfolio tracking.

Academic Tutor—Python & Algorithms

Feb 2023 — Nov 2025

University of Melbourne-Faculty of Engineering & Technology

Melbourne, VIC

- Taught Python programming and algorithmic thinking to cohorts of **200+ students**.

- Designed interactive workshops that improved pass rates by **18% above the class average**, focusing on clarity, structure, and applied understanding.

Co-Founder, Chief Technology Officer

Feb 2023 — Nov 2025

G7 Smart Logistics

Udaipur, India

- Built and Scaled logistics SaaS to **9,000+ daily bookings**, serving **10,000+ truckers** with low digital literacy.
- Engineered the end-to-end software architecture using Java, Kotlin (Android Native), Python (Django) and MySQL.

Education

The University of Melbourne

Melbourne, VIC

Bachelor of Science, Major in Data Science

Jul 2022 — Jun 2025

- **Academic Excellence:** Graduated with a WAM of 85.2/100 (H1 Honor Standard); recognized as a Dean's Honours Scholar (top 3% of cohort) and selected as an International Undergraduate Scholarship recipient.
- **Core Engineering Coursework:** Artificial Intelligence, Machine Learning, Database Systems, and Statistics.
- **Campus Leadership:** Held executive roles including **Education Director** at the Data Science Students Society and **Marketing Director** at GDSC; managed cross-functional teams of 4–5 to deliver technical workshops, hackathons, and industry networking events.

Projects

Autonomous Booking Pipeline Engine

- Designed and deployed a highly scalable serverless pipeline that extracts, normalizes, and actions **13k+ logistics bookings daily** from unstructured WhatsApp chat streams.
- Orchestrated real-time entity extraction utilizing **Gemini** to isolate routing attributes (pickup/drop locations, vehicle specifications) with automated **geocoding integration**.
- Minimized operational latency by hosting the entire ingestion-to-dispatch pipeline on **Cloudflare's edge network**, eliminating heavy server infrastructure overhead.

Generative AI Legal Case Visualizer

- Developed a full-stack educational platform transforming complex legal case studies into interactive graphic novels, deployed on high-performance **Cloudflare** edge infrastructure.
- Engineered a multi-step AI orchestration pipeline using **Gemini Models**; designed a workflow that first generated character schemas to serve as immutable context for subsequent panel generation, successfully solving character consistency.
- Tech Stack: Next.js, Hono, Gemini API, Supabase, Cloudflare (Workers/Pages).

Rental Market Forecasting & Investment Strategy

- Engineered a robust predictive pipeline by scraping and integrating disparate real estate and socioeconomic datasets, utilizing **XGBoost** and regression models to achieve **90%+ R² accuracy**.
- Developed an interactive, investor-centric dashboard to visualize the trade-off between affordability and livability, directly aiding in strategic property acquisition decisions.
- Tech Stack: Python, Pandas, Scikit-learn, Beautiful Soup, Streamlit. [Github Repository](#)

NYC Taxi Fleet Optimization

- Built a distributed Machine Learning model using **PySpark** on a large-scale dataset to predict zone-wise taxi demand under surge conditions.
- Formulated a dynamic fleet allocation strategy that projected a **15% increase in revenue per trip** by optimizing driver distribution based on predicted demand hotspots.
- Tech Stack: PySpark, Apache Spark, MLlib, Cloud Computing. [Github Repository](#)

Certifications & Coursework

- **MLOps Deployment and Life Cycling** via DataCamp (Nov 2025)
- **Machine Learning-Time Series Modelling** via Kaggle (Feb 2024)
- **Marketing Fundamentals** via Netsavvies Media Co. (Oct 2021)
- **Blockchain A-Z:** Learn how to build your first blockchain via Udemy (May 2021)
- **Introduction to Computational Thinking and Data Science** via MITx (Jun 2020)